

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1703

COURSE OUTCOME COVERED

CO2: Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 5

Duration: 1 Hour
Total Marks:50

Annexure A

Scenario Based Assignment

Code of Conduct:

I D. VAISHNAVI (192524143) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Scenario Based Assignment

1. Scenario : A government deploys AI-powered drones to deliver emergency medicines in a flood-affected region. The environment is highly dynamic—roads may suddenly become inaccessible, and weather conditions affect travel cost. The drone has access to: A heuristic estimate (straight-line distance), Partial real-time updates about blocked paths and Variable movement costs depending on terrain and weather. However, due to urgency, the drone must quickly decide routes with minimum delay and risk.

Tasks:

- i. **Explain how Greedy Best-First Search would behave in this scenario and discuss its strengths and weaknesses under uncertainty**

Ans:

Greedy Best-First Search (GBFS) selects the next path based only on the heuristic value $h(n)$, which estimates the remaining distance to the goal (straight-line distance). It ignores the actual travel cost.

In this scenario, the drone always chooses the route that appears closest to the destination, allowing it to make decisions quickly.

Strengths:

- Finds a route quickly.
- Uses less memory.
- Suitable for emergency situations where fast decisions are required.

Weaknesses:

- Does not consider actual travel cost, weather, or terrain.
- May choose blocked or risky paths.
- Does not guarantee the shortest or safest route.
- Performance decreases when the heuristic is inaccurate.

- ii. **Explain how A* would handle the same situation, including how it balances cost and heuristic**

Ans:

A* Search uses the evaluation function:

$$f(n) = g(n) + h(n)$$

where:

- $g(n)$ = Actual cost from the start to the current node.
- $h(n)$ = Estimated cost from the current node to the goal.

A* balances both the actual travel cost and the estimated remaining distance. It avoids expensive or dangerous routes while still moving toward the destination.

Advantages:

- Finds the optimal path if the heuristic is admissible.
- Considers terrain, weather, and travel cost.
- Produces safer and more reliable routes.

iii. Analyze the impact of heuristic accuracy on both algorithms**Ans:**

The heuristic function estimates the remaining distance to the goal.

- **Accurate heuristic:** GBFS and A* expand fewer nodes and find solutions faster.
- **Inaccurate heuristic:** GBFS may choose poor or blocked paths because it depends only on the heuristic. A* may take more time but still finds the optimal solution because it also considers the actual path cost.

Therefore, A* is more reliable than GBFS when heuristic estimates are imperfect.

iv. Discuss how dynamic changes (blocked paths) affect search performance**Ans:**

In a flood-affected region, roads may suddenly become blocked.

- GBFS may continue toward blocked paths because it only follows the heuristic.
 - A* recalculates the path using updated costs and searches for a safer alternative.
 - Frequent changes increase computation time, but A* adapts better to dynamic environments than GBFS.
- v. Recommend the most suitable algorithm and justify your decision considering real-time constraints, optimality, and safety**

Ans:**Recommended Algorithm: A*****Justification:**

- Balances actual travel cost and heuristic information.
- Finds the optimal and safest route.
- Adapts to blocked paths and changing weather.
- Provides reliable decisions for emergency medicine delivery.

Conclusion: A* is the most suitable algorithm because it offers the best balance between real-time performance, optimality, and safety in a dynamic environment

2. Scenario: A smart city implements an AI system to optimize traffic signal timings across hundreds of intersections. The objective is to minimize Total waiting time, Fuel consumption and Traffic congestion. The system must continuously adjust signals based on traffic flow. However the search space is extremely large, traffic patterns change dynamically and the system may get stuck in locally optimal solutions

Tasks. Formulate this as an optimization problem and explain why traditional search is inefficient

Ans:

Optimization Problem:

- **Objective:** Minimize total waiting time, fuel consumption, and traffic congestion.
- **Decision Variables:** Traffic signal timings at each intersection.
- **Constraints:** Traffic rules, pedestrian crossing times, emergency vehicle priority, and synchronization between signals.

Why Traditional Search is Inefficient:

- The number of possible signal timing combinations is extremely large.
- Traffic conditions change continuously.
- Traditional search explores many unnecessary states, making it slow and unsuitable for real-time traffic control.

i. Explain how Hill Climbing would work in this scenario and identify its limitations

Ans:

Hill Climbing starts with an initial traffic signal schedule and continuously changes the timings to improve traffic flow. At each step, it selects the neighboring solution that gives the greatest improvement.

Limitations:

- Can get stuck at a **local maximum**.
- Cannot move through flat regions (**plateaus**).
- May fail to find the global best solution.
- Performance depends on the starting solution.

ii. Discuss issues like local maxima, plateaus, and ridges with real-world interpretation

Ans:

- **Local Maximum:** The system finds a traffic signal plan that seems good but is not the best overall.
- **Plateau:** Several traffic signal settings produce the same performance, so the algorithm cannot determine which direction to move.
- **Ridge:** Better solutions exist, but they require multiple coordinated changes that Hill Climbing cannot easily make.

These situations reduce the efficiency of Hill Climbing in real-world traffic management.

iii. Explain how Simulated Annealing or Genetic Algorithms can overcome these limitations

Ans:

Simulated Annealing:

- Occasionally accepts worse solutions to escape local maxima.
- Gradually reduces random exploration and converges to a better solution.

Genetic Algorithm (GA):

- Maintains multiple traffic signal solutions (population).
- Uses **selection, crossover, and mutation** to generate improved solutions.
- Explores a wider search space and is less likely to get trapped in local optima.

- iv. **Recommend the best approach and justify your choice based on scalability and adaptability**

Ans:

Recommended Approach: Genetic Algorithm (GA)

Justification:

- Handles very large search spaces efficiently.
- Adapts to changing traffic conditions.
- Finds near-optimal solutions without getting trapped in local maxima.
- Suitable for large smart city traffic systems with many intersections.

Conclusion

Genetic Algorithm is the best choice because it is scalable, adaptive, and capable of optimizing complex traffic signal systems in real time.

3. Scenario: An autonomous rover is deployed on Mars to explore unknown terrain. It must: navigate safely to collect samples, avoid hazards (craters, rocks), operate with limited sensing capability and make decisions without a complete map. The rover updates its knowledge as it explores, making this a real-time decision-making problem.

Tasks:

- i. **Explain why this problem requires an online search agent instead of offline search**

Ans:

An **online search agent** is required because the rover does not have a complete map of the environment before starting its mission. It must explore, collect information, and make decisions while moving.

Why online search is suitable:

- The environment is unknown.
- The rover receives information only as it explores.
- It can change its path based on newly discovered obstacles.
- It supports real-time decision making.

Offline search is not suitable because it requires complete knowledge of the environment before planning.

- ii. **Analyze the challenges of partial observe ability and dynamic environments**

Ans:

Partial Observability:

- The rover cannot see the entire terrain.
- Hidden rocks or craters may appear during exploration.
- Limited sensors provide incomplete information.

Dynamic Environment:

- Terrain conditions may change unexpectedly.
- New obstacles may be detected.
- The rover must continuously update its decisions.

These challenges make navigation difficult and require continuous adaptation.

iii. Describe how the rover builds and updates its internal model

Ans:

The rover uses its sensors to collect information about the surroundings.

It:

- Detects rocks, craters, and safe paths.
- Stores newly discovered information in its internal map.
- Updates the map whenever new data is received.
- Uses the updated model to plan safer future movements.

Thus, the rover continuously improves its knowledge while exploring.

iv. Compare online search with uninformed and informed search strategies

Ans:

Online Search	Uninformed Search	Informed Search
Works in unknown environments	Requires a known search space	Uses heuristic information
Learns while exploring	Does not use heuristics	Uses heuristics to guide the search
Adapts to new information	Explores systematically	More efficient than uninformed search
Best for real-time problems	Less suitable for unknown environments	Suitable when a map is available

v. Justify the most suitable approach considering uncertainty, adaptability, and safety

Ans:

Recommended Approach: Online Search Agent

Justification:

- Handles unknown and changing environments.
- Adapts to newly discovered obstacles.
- Updates its internal map continuously.
- Makes safe decisions using real-time sensor information.
- Ensures reliable navigation and sample collection.

Conclusion

An **Online Search Agent** is the most suitable approach because it can adapt to uncertainty, update its knowledge continuously, and safely navigate an unknown Martian environment.

4. Scenario: A university is designing an AI system to automatically generate exam timetables. The system must satisfy multiple constraints: No student should have overlapping exams, Limited number of exam halls, Certain subjects must be scheduled before others, Faculty availability constraints, Special accommodations for some students, The complexity increases as the number of courses and students grows.

Tasks :

Formulate the problem as a Constraint Satisfaction Problem (CSP)

i. **Clearly define variables, domains, and constraints with explanation**

Ans:

A **Constraint Satisfaction Problem (CSP)** consists of variables, domains, and constraints.

- **Variables:** Exams (e.g., Math, Physics, AI, Python).
- **Domains:** Available time slots and exam halls.
- **Constraints:**
 - No student should have overlapping exams.
 - Limited number of exam halls.
 - Some subjects must be conducted before others.
 - Faculty must be available during the assigned time.
 - Special accommodations must be provided for eligible students.

The objective is to assign each exam to a valid time slot and hall while satisfying all constraints.

ii. **Explain how backtracking search works in this scenario**

Ans:

Backtracking Search assigns exams one by one to available time slots and halls.

- If a valid assignment satisfies all constraints, it proceeds to the next exam.
- If a conflict occurs (such as overlapping exams or unavailable halls), it returns to the previous assignment and tries another option.
- This process continues until a complete valid timetable is found.

Backtracking guarantees a valid solution if one exists.

iii. **Discuss how constraint propagation (e.g., forward checking) improves efficiency**

Ans:

Forward Checking removes invalid choices from the remaining unassigned exams after each assignment.

For example:

- If AI is scheduled on Monday at 10 AM, other exams involving the same students or faculty cannot use that time slot.

Advantages:

- Detects conflicts early.
- Reduces the search space.
- Decreases unnecessary backtracking.
- Produces solutions faster.

iv. **Analyze real-world challenges and justify the best strategy for scalability**

Ans:

Real-world Challenges:

- Large number of courses and students.
- Limited exam halls and faculty availability.
- Last-minute schedule changes.
- Special accommodations for some students.
- Increasing complexity as the university grows.

Best Strategy:

Use **Backtracking Search with Constraint Propagation (Forward Checking)**.

Justification:

- Efficiently handles multiple constraints.
- Reduces unnecessary searches.
- Easily scales to large universities.
- Produces valid and conflict-free exam timetables.

Conclusion

Backtracking Search combined with Forward Checking is the most suitable approach because it efficiently solves complex scheduling problems while satisfying all constraints and improving scalability.

5. Scenario: Strategic Game AI for Real-Time Decision Making (Minimax & Alpha-Beta Pruning) A gaming company is developing an AI opponent for a real-time strategy game. The AI must compete against human players, Make decisions within strict time limits, Evaluate complex game states with multiple possible moves, The decision tree is extremely large, making computation expensive.

Tasks:

- i. **Explain how the Minimax algorithm models this problem**

Ans:

Answer:

The **Minimax algorithm** models the game as a decision tree with two players:

- **MAX** (AI player) tries to maximize its score.
- **MIN** (Human player) tries to minimize the AI's score.

The algorithm evaluates all possible moves and selects the move that provides the best outcome, assuming both players play optimally.

- ii. **Analyze the computational challenges of large game trees**

Ans:

The main challenges are:

- The number of possible moves grows rapidly (large search tree).
- Evaluating every game state requires high computation time.
- Large memory is needed to store game states.
- Real-time decision making becomes difficult because of strict time limits.

iii. Explain how Alpha-Beta pruning improves efficiency with detailed reasoning

Ans:

Alpha-Beta Pruning is an optimization technique for the Minimax algorithm.

It removes branches that cannot affect the final decision.

- **Alpha (α):** Best value found so far for the MAX player.
- **Beta (β):** Best value found so far for the MIN player.

If a branch cannot produce a better result than the current alpha or beta value, it is **pruned (ignored)**.

Advantages:

- Reduces the number of nodes evaluated.
- Increases search speed.
- Produces the same optimal result as Minimax.
- Makes real-time game decisions possible.

iv. Design an evaluation function and justify the factors considered

Ans:

An evaluation function assigns a score to each game state.

Example Evaluation Function:

Evaluation Score = (Resources + Army Strength + Territory Control) – (Enemy Strength + Threat Level)

Factors Considered:

- Number of resources available.
- Strength of the AI's army.
- Territory or map control.
- Enemy strength.
- Risk or threat from opponents.

These factors help the AI choose the strongest possible move.

v. Recommend a strategy for balancing optimality and real-time performance

Ans:

Recommended Strategy: Minimax with Alpha-Beta Pruning

Justification:

- Finds the best possible move.
- Eliminates unnecessary branches.
- Reduces computation time.
- Handles large game trees efficiently.
- Suitable for real-time strategy games where quick decisions are required.

Conclusion

Minimax with Alpha-Beta Pruning is the best approach because it provides optimal decisions while reducing computation time, making it ideal for real-time game AI.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand